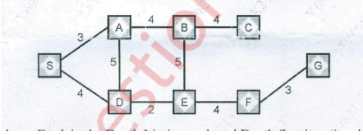
Assignment No.2

1. Define Heuristics. Explain the significance of Heuristic function in the informed search with suitable example.
2. Explain A*algorithm and write its pseudo code.
3. Explain “Simple Reflex based Agent” with the help of schematic diagram or pseudo code.
4. Apply greedy best-first search. At each iteration, each node is expanded using evaluation function f (n)= h(n). h (S) = 10, h (A) = 10, h(D) = 8, h(B) = 6, h(E) = 6.5, h (C) = 4, h (F) = 3, h(G) = 0. S is start state and G is goal state.



5. Explain the Depth Limit search and Depth first iterative deepening search.
6. Formulate the problem; choose the formulation that is precise enough to be implemented. Also identify the initial state, goal test, successor function, and cost function for the following. Problem statement: Autonomous Taxi driver
7. Explain Goal based Agent with a block diagram
8. Utility based Agent with a block diagram.
9. Formulate the vaccum world,
10. 8 queens problem
11. 8 puzzle problem.
12. List uninformed search strategies explain any two of them.
13. Explain memory bounded search
14. Explain SMBA* search.
15. Apply BFS, DFS, and Iterative Deepening search to Romania.
16. Explain RBFS in detail.